

Mohamed Khalil Zrelly

Generative AI Engineer · Agentic Systems · RAG Pipelines · LLM Architectures

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| [makiz-code.github.io/MKirell](https://github.com/makiz-code)

Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, designing agentic systems, RAG pipelines, and LLM-powered workflows. Engineering student at Sup Galilée — Sorbonne Paris North University, with a background in machine learning, deep learning, and natural language processing.

Work Experience

Crédit Agricole Personal Finance & Mobility

Generative AI Engineer Apprentice

France
September 2025 — Present

Developed a declarative, config-driven **graph workflow orchestration** microservice, enabling dynamic workflow composition across a scalable and maintainable AI ecosystem.

Built an automated **Red Teaming** framework for AI workflows, probing LLM behaviour across guardrails and targeted adversarial scenarios to strengthen model resilience.

Agentic Workflows · Advanced RAG · AWS Bedrock · AI Guardrails · LLM Observability · Python/FastAPI · Amazon DynamoDB

VERMEG for Banking & Insurance Software

Natural Language Processing Intern

Tunisia
February 2024 — June 2024 · 5 months

Implemented a system automating **email classification** and information extraction from attachments using **BERT** and **OCR** to optimise efficiency and processing time.

Developed a web application for **BERT fine-tuning**, email management, and real-time data analysis and visualisation to enhance user interaction and performance monitoring.

Hugging Face Transformers · Fine-Tuning · NER · PyTorch · Tesseract OCR · Python/Flask · React/Redux · MongoDB

Sofiatech

Machine Learning Intern

Tunisia
June 2023 — August 2023 · 3 months

Evaluated multiple **deep reinforcement learning** algorithms from Stable Baselines 3 across various Gymnasium environments to assess agent performance and training efficiency.

Built a **2D grid game** in Pygame where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas

Featured Projects

CVision | CV Recommendation System

Jun–Aug 2025 | github.com/makiz-code/CVision

Built a **RAG**-based system that optimises automatic selection and ranking of the most relevant candidate profiles for a given job description.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Weaviate · Python/Streamlit

Twixel | Multi-Agent Social Listening System

In Progress

Designing an intelligent **multi-agent system** for automated social listening on X (Twitter), identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · Milvus · Python/FastAPI · Vue.js

Technical Skills

Generative AI & Agentic | LangChain, LangGraph, AWS SageMaker, AWS Bedrock

ML / DL / RL | PyTorch, TensorFlow, Keras, Scikit-Learn, Stable Baselines 3, Gymnasium

NLP & LLMs | Hugging Face Transformers, Sentence-Transformers, spaCy, Fine-Tuning, RAG

Data & Big Data | Pandas, NumPy, Matplotlib, Apache Airflow, PySpark

Databases | PostgreSQL, MongoDB, Amazon DynamoDB, MySQL, Oracle

Web & APIs | React, Vue.js, FastAPI, Flask, Express.js, Laravel

DevOps & Cloud | AWS, Docker, Kubernetes, Terraform, GitHub Actions

Programming Languages | Python, JavaScript/TypeScript, Java, C/C++, PHP, PL/SQL

Education

Sup Galilée – Sorbonne Paris North University

Engineering Degree in Computer Science

Villetaneuse, France
2024 — 2027 (Expected)

ISI Ariana – University of Tunis El Manar

Bachelor of Science in Computer Science

Ariana, Tunisia
2021 — 2024 · High Distinction

Certifications & Languages

Microsoft Azure | Azure AI Fundamentals AI-900 (Jan 2024) · Azure Data Fundamentals DP-900 (Jan 2024)

Udemy | Spark and Python for Big Data with PySpark (Mar 2024) · NLP with Python (Feb 2024)

Spoken Languages | Arabic (Native) · French B2 — TCF · English B2 — TOEIC